

BOTOSANI, Romania

WMO: 150200

Lat: 47.7356N

Lon: 26.6456E

Elev: 532

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	8.6	-2.2	5.0	3.3	3.8	6.8	10.4	24.9	34.5	21.0	30.5	3.1	300	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	90.4	69.8	87.1	68.7	84.1	67.3	72.5	84.5	70.9	82.6	69.4	80.5	5.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.9	108.3	77.4	67.2	102.0	75.6	65.7	96.6	73.8	36.5	84.1	35.1	82.8	33.7	80.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.4	14.6	-4.7	95.2	7.1	3.9	-9.8	98.1	-14.0	100.4	-18.0	102.6	-23.1	105.5		
(5)			-5.1	75.5	7.0	1.9	-10.2	76.9	-14.3	78.0	-18.2	79.1	-23.2	80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	28.0	31.7	40.1	51.5	61.3	67.6	70.9	69.8	60.8	50.1	40.8	30.9
	DBStd	17.12	10.24	10.79	8.87	7.28	6.60	5.67	5.09	5.56	6.65	7.78	8.71	9.88
	HDD50	2576	684	515	320	66	4	0	0	0	3	93	298	593
	HDD65	5875	1148	933	771	406	149	38	10	17	156	463	727	1057
	CDD50	2723	0	2	14	112	356	527	649	614	329	98	21	1
	CDD65	548	0	0	0	2	36	115	195	166	31	3	0	0
	CDH74	5348	0	0	0	42	421	1097	1796	1610	353	29	0	0
	CDH80	1788	0	0	0	7	89	329	647	622	91	3	0	0
Wind	WSAvg	5.6	5.6	6.1	6.8	6.6	5.8	5.5	5.1	4.6	5.1	5.1	5.8	5.4
Precipitation	PrecAvg	23.3	0.8	0.9	1.2	2.1	2.6	3.3	3.8	2.5	1.9	1.8	1.2	1.1
	PrecMax	31.7	2.0	1.9	2.8	5.9	6.1	6.8	8.4	7.1	6.0	7.4	2.7	3.0
	PrecMin	13.9	0.2	0.2	0.3	0.3	0.5	0.7	0.4	1.0	0.2	0.3	0.0	0.0
	PrecStd	5.0	0.4	0.5	0.7	1.3	1.6	1.6	2.2	1.6	1.5	1.5	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	59.8	68.7	79.0	86.6	90.4	94.5	95.3	87.2	77.6	67.7	56.0
		MCWB	44.4	49.2	52.5	59.7	64.8	69.8	69.4	71.3	66.4	62.1	55.7	47.5
	2%	DB	47.8	53.4	62.7	73.3	82.3	86.9	90.3	90.7	82.2	71.7	61.3	50.4
		MCWB	41.1	45.7	50.2	57.5	63.5	69.0	70.0	70.0	64.6	59.5	53.3	44.6
	5%	DB	43.6	48.6	57.9	69.2	78.8	83.8	86.7	86.8	77.9	66.8	56.8	46.4
		MCWB	38.9	42.7	47.5	55.4	62.8	67.3	69.2	68.9	63.0	57.2	50.9	42.0
	10%	DB	40.6	44.8	53.7	65.2	75.1	80.7	83.3	82.8	73.5	62.5	53.2	42.7
		MCWB	37.2	40.3	45.2	53.1	60.8	66.3	68.0	67.6	61.6	54.8	49.0	39.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	50.0	53.8	62.2	68.1	73.8	74.5	74.9	69.6	64.1	57.0	49.0
		MCDB	51.8	57.4	66.4	76.1	80.1	84.5	85.9	88.0	83.8	75.3	65.5	54.3
	2%	WB	42.0	46.3	50.8	59.2	65.7	71.5	72.4	72.4	66.5	60.8	54.1	45.3
		MCDB	46.5	52.8	61.0	71.4	77.7	82.3	84.2	85.1	77.2	69.9	59.8	49.9
	5%	WB	39.5	43.4	48.5	56.7	63.9	69.6	70.9	70.5	64.5	58.2	51.8	42.4
		MCDB	43.0	47.9	56.8	66.4	75.8	79.7	82.6	82.6	74.5	65.0	56.1	46.0
	10%	WB	37.3	40.6	46.0	54.3	62.1	67.9	69.3	68.7	62.6	55.7	49.3	39.7
		MCDB	40.2	44.5	52.5	63.0	73.0	77.7	80.4	80.0	71.7	61.8	53.0	42.5
Mean Daily Temperature Range	5% DB	MDBR	10.7	12.3	15.7	18.8	20.7	19.8	20.5	21.1	19.3	16.8	12.1	10.7
		MCDBR	14.9	18.5	23.7	26.0	27.1	25.2	26.8	28.1	27.7	24.7	18.8	16.0
		MCWBR	10.3	12.0	13.4	12.9	11.7	10.2	10.0	10.5	11.8	13.5	11.5	11.2
	5% WB	MCDBR	13.5	17.5	22.4	23.6	24.1	21.8	23.0	24.4	23.2	22.3	16.8	15.0
	MCWBR	9.6	11.7	13.1	12.1	11.3	10.3	9.8	10.0	10.9	12.5	10.6	10.9	
Clear-Sky Solar Irradiance	taub	0.319	0.350	0.387	0.447	0.419	0.417	0.446	0.452	0.408	0.373	0.345	0.317	
	taud	2.278	2.229	2.212	2.086	2.241	2.279	2.203	2.179	2.259	2.336	2.352	2.330	
	Ebn at Noon	234	252	260	255	268	269	259	251	251	241	223	220	
	Edh at Noon	24	32	38	48	42	41	43	43	36	29	23	21	
All-Sky Solar Radiation	RadAvg	361	603	990	1408	1791	1885	1872	1684	1177	747	365	275	
	RadStd	53	89	105	125	145	158	134	124	146	95	33	37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.22	N/A	N/A	N/A	N/A	N/A	N/A	-326	+176	N/A
(47) Regional (31 neighbors)		+1.24	N/A	N/A	+1.51	N/A	N/A	-263	-371	+185	+61

Nomenclature: See separate page